



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2016
Higher 2

CANDIDATE
NAME

--

CIVICS
GROUP

1	5	S			
----------	----------	----------	--	--	--

INDEX
NUMBER

--	--

H2 BIOLOGY

9648/03

Applications Paper and Planning Question

20 September 2016

Paper 3

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough workings.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

At the end of the examination,

1. Fasten all your work securely together.
2. Hand in the following separately:
 - Structured questions 1 – 3
 - Planning question
 - Essay question

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use	
1	/ 13
2	/ 13
3	/ 14
4 Planning	/ 12
5 Essay	/ 20
Total	/ 72

This paper consists of **12** printed pages.

[Turn over]

QUESTION 1

ABL1 is a proto-oncogene that encodes a protein tyrosine kinase involved in a variety of cellular processes, including cell division. A researcher intended to mass produce *ABL1* protein tyrosine kinase using bacterial cells. He obtained *ABL1* cDNA from the cDNA library and the complete sequence of *ABL1* cDNA non-template strand is shown in **Fig. 1.1**. The start and stop triplets are bolded.

```

5' 1 CCTATTACTTTATGGGGCAGCAGCCTGGATCCGT-----
    61 -----
    121 -----GCATCTGACTTTG
    181 AGCCTCAGGG TCTGAGTGAAGCTTCT 3'
    
```

Fig. 1.1

- (a) Explain why the gene of interest is obtained from cDNA library instead of genomic DNA library. [2]

.....

.....

.....

.....

The plasmid pUC18, was first chosen for the cloning of *ABL1* cDNA. **Fig. 1.2a** shows a diagram of the plasmid pUC18 and the position of the restriction sites found in the plasmid.

Fig. 1.2b shows two restriction sites commonly used in genetic engineering.

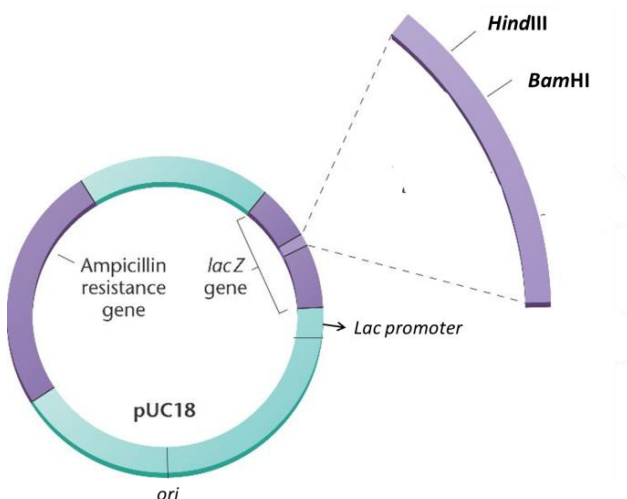


Fig. 1.2a

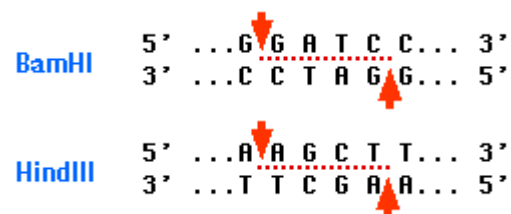


Fig. 1.2b

(b) With reference to **Fig. 1.1** and **Fig. 1.2**, comment on the effectiveness of using pUC18 for *ABL1* protein production. [3]

.....

.....

.....

.....

.....

.....

The researcher found that the *ABL1* cDNA is present in low copy number in the cDNA library. Hence, he carried out PCR to amplify the *ABL1* cDNA.

(c) Suggest two reasons why the total amount of amplified *ABL1* cDNA did not increase between the 30th cycle and the 40th cycle. [2]

1.
2.

An alternative cloning technique called **TA** cloning shown in **Fig. 1.3** can also be used to clone the *ABL1* cDNA. The *Taq* DNA polymerase used in PCR has a non-template dependent activity which preferentially adds a single adenine nucleotide to the 3'-ends of a double stranded DNA molecule. This results in PCR product with 3'-A overhangs. This enables the *ABL1* cDNA to be inserted into a plasmid designed to have 3'-T overhangs.

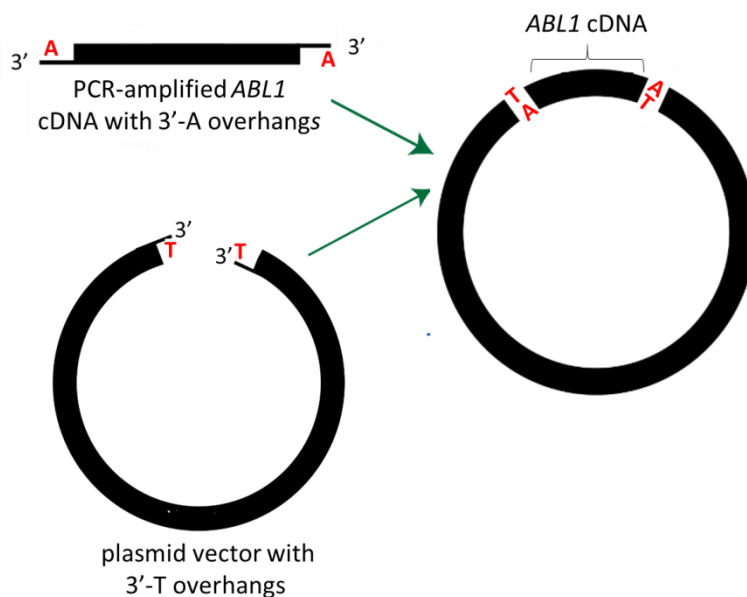


Fig. 1.3

(d) (i) With reference to **Fig. 1.3**, describe how the PCR-amplified *ABL1* cDNA can be inserted into the plasmid using **TA** cloning technique. [4]

.....

.....

.....

.....

.....

.....

.....

.....

(ii) Suggest one advantage and one disadvantage of using **TA** cloning. [2]

Advantage:

.....

.....

Disadvantage:

.....

.....

[Total: 13]

QUESTION 2

Plants have developed defence mechanisms against pathogens such as bacteria, fungi and viruses. Chemicals released by these pathogens can trigger a defence response in infected plant cells. For example, the production of hydrogen peroxide (H_2O_2) which reacts with pathogen membranes and cellular chemicals eventually kills both the cell and pathogen.

The *OSRac1* gene from another plant species was isolated and introduced into a number of rice plant (*Oryza spp.*) lines to study its role in disease resistance of plants to blast fungus. Experiments were carried out to see if the *OSRac1* gene was part of the signalling pathway for hydrogen peroxide production. A control (C) and four other genetically modified rice plant lines (A1, A2, D1 and D2) grown *in vitro* from calluses were exposed to chemicals known to initiate a defence response by producing hydrogen peroxide. A1 and A2 are rice plants with the *OSRac1* gene always turned on. D1 and D2 are rice plants with the *OSRac1* gene suppressed. The results are shown in the Fig. 2.1.

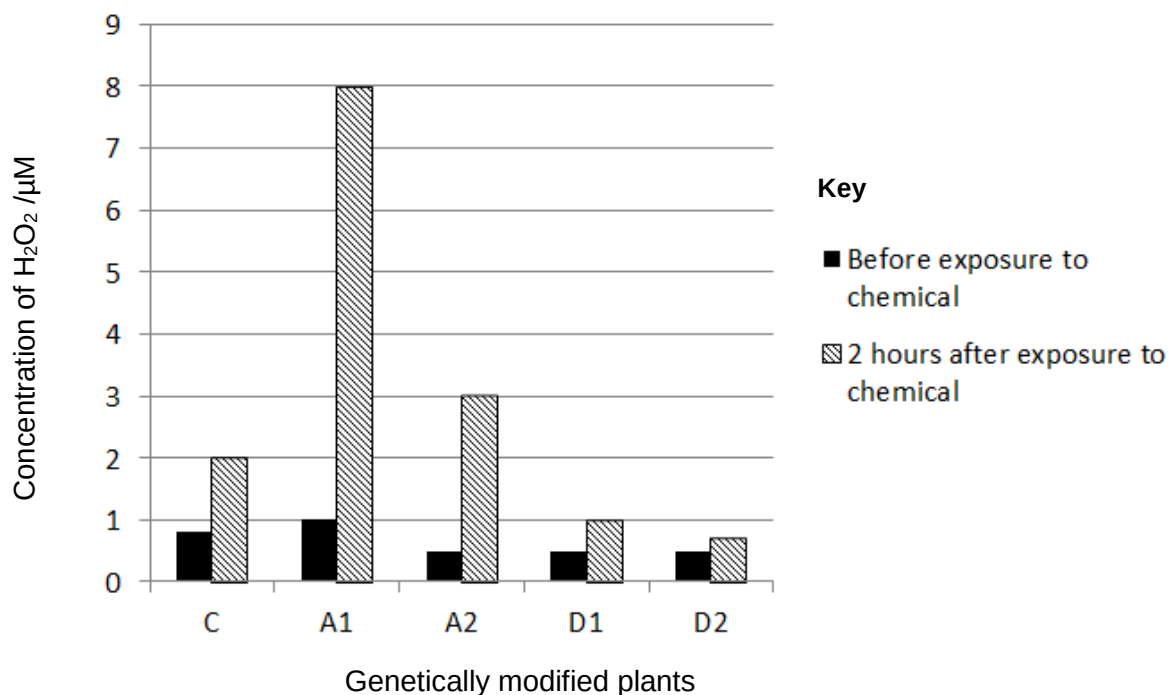


Fig. 2.1

(a) Describe how calluses are obtained from rice plants.

[3]

.....

.....

.....

.....

.....

.....

- (b) With reference to **Fig. 2.1**, compare the **change in H_2O_2 production** between the control and genetically modified plants two hours after the chemical was applied. [2]

.....

.....

.....

.....

.....

- (c) Evaluate whether the data supports the hypothesis that *OSRac1* gene is involved in disease resistance. [2]

.....

.....

.....

.....

Rice blast caused by the fungal pathogen is a destructive disease of rice. The use of blast resistance genes is an effective way to control the fungal disease in rice and to reduce losses in crop yield.

Recently, researchers identified a known genetic marker that is tightly linked to the blast resistance genes in some fungal resistant crops. This allows the identification of crops with blast resistance and subsequent cloning of transgenic rice with the blast resistance gene.

In a rice breeding programme, researchers wanted to identify the blast resistant crops from those that are susceptible to blast.

- (d)** Using the information provided, describe how RFLP analysis can help to distinguish between blast resistant crops and those that are susceptible to blast. [6]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[Total: 13]

QUESTION 3

(a) Describe how the normal copy of a gene can be introduced to a patient's cells via non-viral method. [3]

.....

.....

.....

.....

.....

.....

Type 1 diabetes is a condition where an individual's Islet cells cannot produce insulin in response to high blood glucose levels. Patients usually are dependent on insulin injections.

Transplanting Islets from donors has been studied as a form of treatment for Type I diabetes for over three decades.

Islet cells are usually encapsulated in alginate microspheres before transplanting them into patients. The alginate microsphere creates a barrier between the donor cells and the recipient's cells.

Fig. 3.1 shows some Islet cells encapsulated in an alginate microsphere.

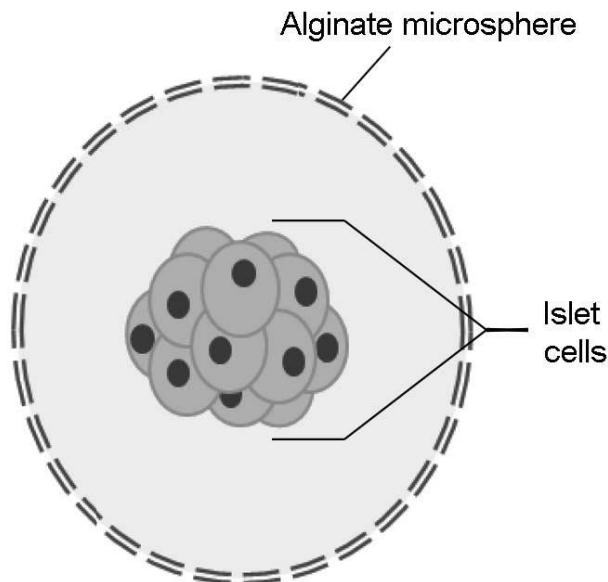


Fig. 3.1

(b) Suggest why Islet cells were encapsulated before they were transplanted into a patient. [1]

.....

.....

Patients transplanted with human Islet cells obtained from deceased individuals can be made insulin independent for around 5 years using the Islet encapsulation treatment. However, this approach is limited because of the scarcity and quality of donor Islet cells.

(c) Suggest why Islet encapsulation treatment lasts only approximately 5 years. [2]

.....

.....

.....

.....

(d) Researchers have also looked into producing human insulin-producing beta cells from stem cells.

A recent study in 2014 used a human embryonic stem cell line (HUES8) and a human induced pluripotent stem cell (hiPSC) line to develop two types of human insulin-producing beta cells called HUES8 SC- β and hiPSC SC- β respectively to overcome the problem of scarcity.

The human induced pluripotent stem cells were derived from fully differentiated adult somatic cells as shown in **Fig. 3.2**.

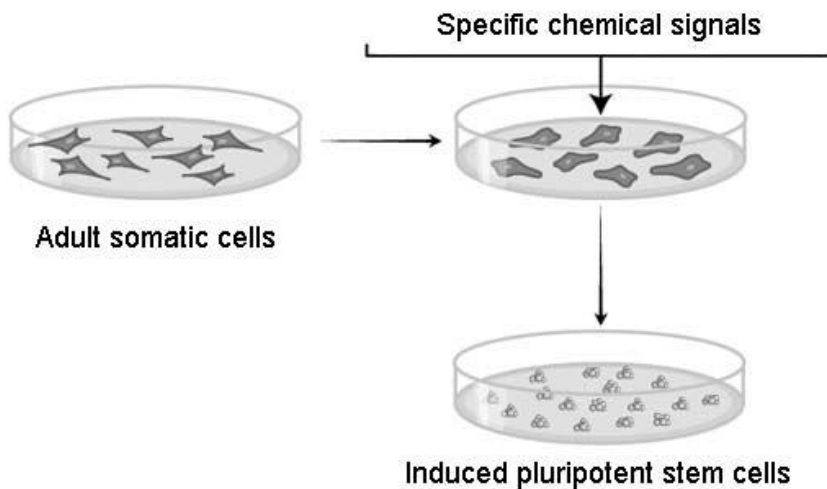


Fig. 3.2

(i) Explain why fully differentiated somatic cells can be induced to become pluripotent. [2]

.....

.....

.....

.....

- (ii) State one ethical issue related to stem cell research and explain how using induced pluripotent stem cells would address this issue. [2]

.....

.....

.....

.....

- (e) Trials have been done by transplanting HUES8 SC- β cells and hiPSC SC- β cells into diabetic mice using alginate microsphere encapsulation. Human insulin produced by the mice was measured following a high carbohydrate meal.

The results of this study were recorded in **Table 3.1**.

Type of beta cell transplanted into diabetic mice	Mean concentration of human insulin secreted $\pm$ standard deviation / μg per ml of blood
HUES8 SC- β cells	2.3 ± 0.2
hiPSC SC- β cells	2.2 ± 0.3
Normal human beta cells	2.1 ± 0.9

Table 3.1

- (i) Compare the secretion of insulin by these three types of transplanted cells. [3]

.....

.....

.....

.....

.....

- (ii) Explain the purpose of the normal human beta cells. [1]

.....

.....

[Total: 14]

QUESTION 4 – Planning question

You are required to plan, but not carry out, an experiment to investigate the effect of temperature on the rate of respiration in mung beans.

Germination of mung beans can be initiated by soaking the seeds overnight. The mung beans can then be placed into a simple respirometer (**Fig. 4.1**) to measure the rate of respiration by measuring oxygen uptake of the seeds. Soda lime pellets absorb any carbon dioxide produced by the germinating seeds. As oxygen is taken up during respiration, the drop of coloured liquid introduced in the capillary tube by capillary action is displaced.

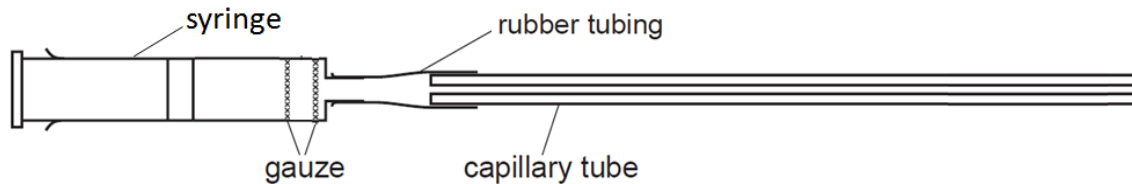


Fig. 4.1: A respirometer

You must use the items from this list:

- 200 mung beans of equal size that have been soaked for 24 hours
- Soda lime pellets
- Syringes
- Rubber tubing connected to glass capillary tube with 1mm bore diameter
- Beaker of coloured liquid
- Ruler marked in mm
- Paper towels
- Stop watch
- Thermostatically controlled incubator
- Other available laboratory apparatus and equipment



Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure that they are as accurate and as reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken

[Total: 12]

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question

QUESTION 5

- (a)** Discuss the detrimental environmental and economic effects of growing genetically-modified herbicide resistant crops. [6]
- (b)** Discuss the ethical and social issues of the Human Genome Project. [6]
- (c)** Describe how a genetic condition like SCID may be treated with viral gene therapy and discuss the potential limitations of this kind of treatment. [8]

[Total: 20]

• END OF PAPER 3 •